Tutorial-6 Solutions

Friday, 25 September 2026 4:02 PM

1) **Given:**

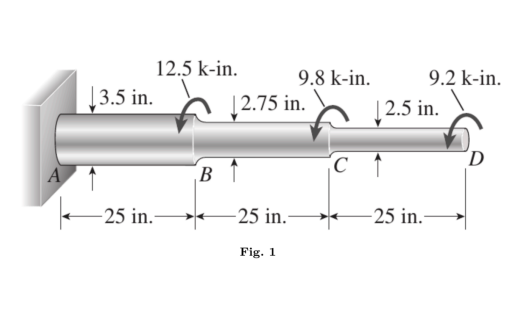


Fig. 1

G (shear modulus) $= 11.6\times10^3$ ksi

(a) Max. shear stress (τ_{max}) in shaft:

Now, in figure, all three torques act in same direction.

Internal torques (T) for each segment is calculated by summing the applied torques from the free end D towards the fixed end D.

$\therefore\ T_{CD} = 9.2$ k-in.

$T_{BC} = (9.2 + 9.8)$ k-in. $= 19$ k-in.

$T_{AB} = (19 + 12.5)$ k-in. $= 31.5$ k-in.

Now, polar moment of inertia in each segment:

$J_{CD} = \frac{\pi}{32}\times(2.5)^4 = 3.835\ in^4$

$J_{BC} = \frac{\pi}{32}\times(2.75)^4 = 5.6147\ in^4$

$J_{AB} = \frac{\pi}{32}\times(3.5)^4 = 14.732\ in^4$

Now, shear stress is given by $\tau = \frac{16T}{\pi d^3}$

$\tau_{CD} = \frac{16\times9.2}{\pi\times(2.5)^3} = 2.998$ ksi

$\tau_{BC} = \frac{16\times(19)}{\pi\times(2.75)^3} = 4.653$ ksi

$\tau_{AB} = \frac{16\times31.5}{\pi\times(3.5)^3} = 3.741$ ksi

$\therefore$ Max. shear stress in shaft $\Rightarrow \max.(\tau_{AB}, \tau_{BC}, \tau_{CD})$

$$\therefore\ \boxed{\tau_{max} = 4.653\ \text{ksi}}$$

(b) Angle of twist at end D (ϕ_D):

$$\phi_D = \sum \frac{T_iL_i}{GJ_i} = \left(\frac{TL}{GJ}\right)_{AB} + \left(\frac{TL}{GJ}\right)_{BC} + \left(\frac{TL}{GJ}\right)_{CD}$$

$$\phi_D = \frac{31.5\times25}{G\times14.732} + \frac{19\times25}{G\times5.6147} + \frac{9.2\times25}{G\times3.835}$$

$\because G = 11.6\times10^3$ ksi

$\therefore\ \phi_D = 0.01707$ rad.

Converting to degrees:

$\phi_D = 0.01707\times\frac{180}{\pi}$

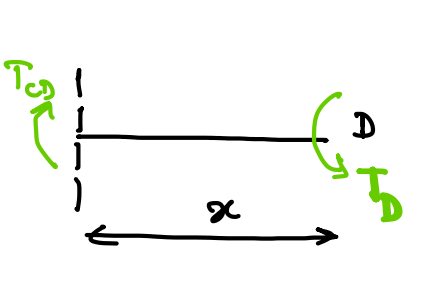
$$\boxed{\phi_D = 0.978^\circ}$$

(c) Angle of twist using energy method:

Strain energy of shaft:

$$U = \sum\frac{T_i^2L_i}{2GJ_i} = \left(\frac{T^2L}{2GJ}\right)_{AB} + \left(\frac{T^2L}{2GJ}\right)_{BC} + \left(\frac{T^2L}{2GJ}\right)_{CD}$$

Now, expressing internal torques:



$T_{CD} = T_D$

Ilrly, $T_{BC} = T_D + 9.8$

$T_{AB} = T_D + 9.8 + 12.5 = T_D + 22.3$

Applying Castigliano's theorem $\Rightarrow$

$$\phi_D = \frac{\partial U}{\partial T_D} = \left[\frac{\cancel{2}T\frac{\partial T}{\partial T_D}\cdot L}{\cancel{2}GJ}\right]_{AB} + \left[\frac{\cancel{2}T\frac{\partial T}{\partial T_D}\cdot L}{\cancel{2}GJ}\right]_{BC} + \left[\frac{\cancel{2}T\frac{\partial T}{\partial T_D}\cdot L}{\cancel{2}GJ}\right]_{CD}$$

$$\phi_D = \frac{T_{AB}L_{AB}(1)}{GJ_{AB}} + \frac{T_{BC}L_{BC}(1)}{GJ_{BC}} + \frac{T_{CD}L_{CD}(1)}{GJ_{CD}}$$

Substitute $T_D = 9.2$ k-in in T_{AB}, T_{BC}, & T_{CD} :-

$$\phi_D = \frac{31.5\times25}{G\times14.732} + \frac{19\times25}{G\times5.6147} + \frac{9.2\times25}{G\times3.835}$$

$$\boxed{\phi_D = 0.978^\circ}$$

② Tapered bar AB with linearly varying diameter.

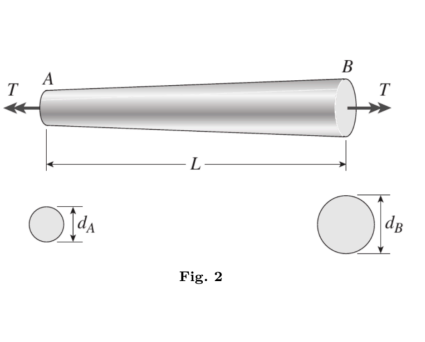
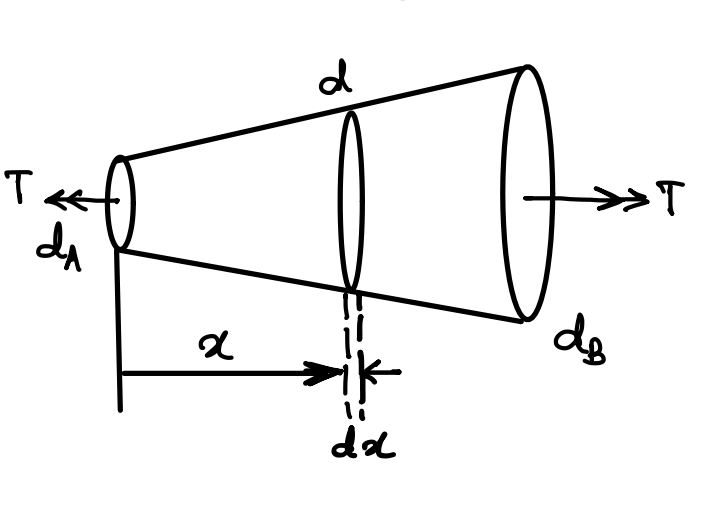


Fig. 2



$$d = d_A + \left(\frac{d_B - d_A}{L}\right)x$$

$$d\phi = \frac{T\,dx}{GJ}$$

$$J = \frac{\pi}{32}d^4 = \frac{\pi}{32}\left(d_A + \left(\frac{d_B - d_A}{L}\right)x\right)^4$$

$$\therefore\ \int_0^\phi d\phi = \int_0^L \frac{T\,dx}{G\times\frac{\pi}{32}\left(d_A + \left(\frac{d_B - d_A}{L}\right)x\right)^4}$$

Let, $d_A + \frac{d_B - d_A}{L}x = u$

Now, $du = \left(\frac{d_B - d_A}{L}\right)dx$

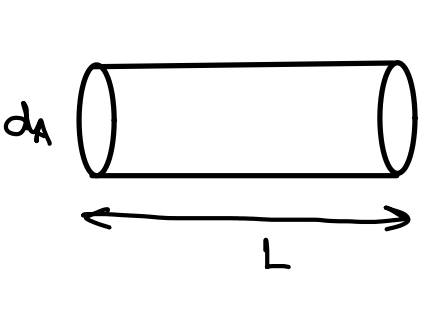
$$\therefore\ \phi = \int_{d_A}^{d_B}\frac{T\,du}{\left(\frac{d_B - d_A}{L}\right)G\times\frac{\pi}{32}u^4}$$

$$\phi = \frac{32TL}{G\pi(d_B - d_A)}\cdot\int_{d_A}^{d_B}\frac{du}{u^4}$$

$$\phi = \frac{32TL}{G\pi(d_B - d_A)}\times\left[\frac{u^{-3}}{-3}\right]_{d_A}^{d_B}$$

$$\phi_{tap} = \frac{32TL}{3G\pi(d_B - d_A)}\times\left[\frac{1}{d_A^3} - \frac{1}{d_B^3}\right] \quad \text{——— ①}$$

Now,



$$\phi_{pris} = \frac{TL}{G\frac{\pi d_A^4}{32}}$$

$$\phi_{pris} = \frac{32TL}{\pi G d_A^4}$$

Given condition $\Rightarrow$

$$\phi_{tap} = \frac{1}{2}\phi_{pris}$$

$$\frac{\cancel{32}\cancel{T}\cancel{L}}{3\cancel{G}\cancel{\pi}(d_B - d_A)}\times\left[\frac{1}{d_A^3} - \frac{1}{d_B^3}\right] = \frac{1}{2}\times\frac{\cancel{32}\cancel{T}\cancel{L}}{\cancel{\pi}\cancel{G}d_A^4}$$

$$\Rightarrow\ \frac{1}{3(d_B - d_A)}\times\left[\frac{d_B^3 - d_A^3}{\cancel{d_A^3}d_B^3}\right] = \frac{1}{2d_A^{\cancel{4}\,1}}$$

$$\Rightarrow\ \frac{d_B^2 + d_Bd_A + d_A^2}{3d_B^3} = \frac{1}{2d_A}$$

$\Rightarrow$ let $\frac{d_A}{d_B} = x$;

$$\frac{1}{3}\left[1 + \left(\frac{d_A}{d_B}\right) + \left(\frac{d_A}{d_B}\right)^2\right] = \frac{d_B}{2d_A}$$

$$\Rightarrow\ \frac{2}{3}\left[1 + x + x^2\right] = \frac{1}{x}$$

$$x^3 + x^2 + x = 3/2$$

$$x^3 + x^2 + x - 3/2 = 0$$

Solving cubic eqn :-

$x_1 = 0.6914$ $\quad x_2$ & x_3 are imaginary roots (reject)

$\therefore\ \frac{d_A}{d_B} = 0.6914$

Hence, $\boxed{\frac{d_B}{d_A} = 1.4463}$

③ In the assembled state both shafts are loaded by the torque M_T.

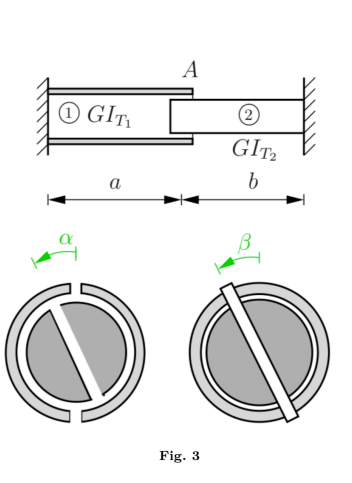


Fig. 3

We cut the system at position A and determine the angle of twist of ① and ② separately:

$\phi_1 = \frac{M_Ta}{GI_{T_1}}$ $\qquad \phi_2 = \frac{M_Tb}{GI_{T_2}}$

From geometric compatibility in assembled state;

$\phi_1 = \alpha - \phi_2$ ——— ①

$\phi_1 = \beta$ ——— ②

$$\frac{M_Ta}{GI_{T_1}} = \alpha - \frac{M_Tb}{GI_{T_2}}$$

$$M_T\left[\frac{a}{GI_{T_1}} + \frac{b}{GI_{T_2}}\right] = \alpha$$

$$\boxed{M_T = \frac{\alpha}{\frac{a}{GI_{T_1}} + \frac{b}{GI_{T_2}}}}$$

$$\beta = \phi = \left(\frac{\alpha}{\frac{a}{GI_{T_1}} + \frac{b}{GI_{T_2}}}\right)\left[\frac{a}{GI_{T_1}}\right]$$

$$\boxed{\beta = \frac{\alpha\cdot a}{a + b\left(\frac{I_{T_1}}{I_{T_2}}\right)}}$$

④ Given:

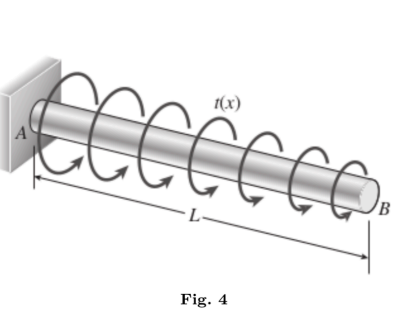
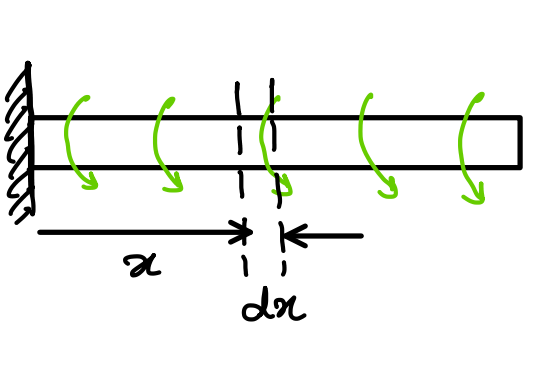


Fig. 4

Intensity of torque $t(x)$ varies linearly from max. value t_A at end A to zero at end B.



$$t(x) = t_A - \left(\frac{t_A}{L}\right)x$$

$$J = \frac{\pi}{32}d^4$$

Torque $= t(x).dx$

(a)

$$\tau_{max} = \int_0^L\frac{16\,t(x).dx}{\pi d^3}$$

$$\tau_{max} = \frac{16}{\pi d^3}\left[\int_0^L\left[t_A - \left(\frac{t_A}{L}\right)x\right]dx\right]$$

$$\tau_{max} = \frac{16}{\pi d^3}\left[t_AL - \frac{t_A}{L}\times\left(\frac{L^2}{2}\right)\right]$$

$$\tau_{max} = \frac{16}{\pi d^3}\left[t_AL - \frac{t_AL}{2}\right]$$

$$\boxed{\tau_{max} = \frac{8t_AL}{\pi d^3}}$$

(b)

$$\phi = \int_0^L\frac{t(x).dx.x}{G\frac{\pi}{32}d^4}$$

$$\phi = \frac{32}{G\pi d^4}\left[\int_0^L\left(t_A - \frac{t_A}{L}x\right)x\,dx\right]$$

$$\phi = \frac{32}{G\pi d^4}\left[t_A\frac{L^2}{2} - \frac{t_A}{L}\times\frac{L^3}{3}\right]$$

$$\phi = \frac{32L^2}{G\pi d^4}\times\frac{t_A}{6}$$

$$\boxed{\phi = \frac{16t_AL^2}{3\pi Gd^4}}$$

⑤ Tube 2 is heat-shrunk onto shaft 1.

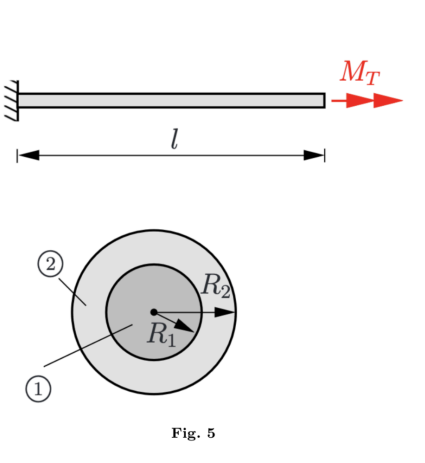


Fig. 5

$\therefore$ They will deform as single unit.

$$\Rightarrow\ \boxed{\phi_1 = \phi_2 = \phi} \quad \text{——— ①}$$

Total external torque applied $\Rightarrow M_T$

$$\boxed{M_T = T_1 + T_2} \quad \text{——— ②}$$

Now, $J_1 = \frac{\pi R_1^4}{2}$ & $J_2 = \frac{\pi}{2}(R_2^4 - R_1^4)$

$$\phi = \frac{T_1l}{G_1J_1} = \frac{T_2l}{G_2J_2}$$

$\therefore\ T_1 = \frac{G_1J_1\phi}{l}$ $\qquad T_2 = \frac{G_2J_2\phi}{l}$

Substituting in ② $\Rightarrow$

$$M_T = \frac{G_1J_1\phi}{l} + \frac{G_2J_2\phi}{l}$$

$$lM_T = (G_1J_1 + G_2J_2)\phi$$

$$\Rightarrow\ \phi = \frac{M_Tl}{G_1\frac{\pi}{2}R_1^4 + G_2\frac{\pi}{2}(R_2^4 - R_1^4)}$$

$$\Rightarrow\ \boxed{\phi = \frac{M_Tl}{\frac{\pi}{2}R_1^4\left[G_1 + G_2\left(\frac{R_2^4}{R_1^4} - 1\right)\right]}}$$

Maximum shear stress:

$$\tau_{1,max} = \frac{T_1R_1}{J_1} = \frac{\left(\frac{G_1\cancel{J_1}}{l}\phi\right)R_1}{\cancel{J_1}}$$

$$\tau_{1,max} = \frac{G_1\phi R_1}{l}$$

$$\tau_{1,max} = \frac{G_1R_1}{\cancel{l}}\times\left[\frac{M_T\cancel{l}}{\frac{\pi}{2}R_1^{\cancel{4}}\left[G_1 + G_2\left(\frac{R_2^4}{R_1^4} - 1\right)\right]}\right]$$

$$\boxed{\tau_{1,max} = \frac{G_1M_T}{\frac{\pi}{2}R_1^3\left[G_1 + G_2\left(\frac{R_2^4}{R_1^4} - 1\right)\right]}}$$

Ilrly, $\tau_{2,max} = \frac{T_2R_2}{J_2} = \frac{\left(\frac{G_2\cancel{J_2}}{l}\phi\right)R_2}{\cancel{J_2}}$

$$\tau_{2,max} = \frac{G_2\phi R_2}{l}$$

$$\tau_{2,max} = \frac{G_2R_2}{\cancel{l}}\times\left[\frac{M_T\cancel{l}}{\frac{\pi}{2}R_1^4\left[G_1 + G_2\left(\frac{R_2^4}{R_1^4} - 1\right)\right]}\right]$$

$$\boxed{\tau_{2,max} = \frac{G_2R_2M_T}{\frac{\pi}{2}R_1^4\left[G_1 + G_2\left(\frac{R_2^4}{R_1^4} - 1\right)\right]}}$$

⑥ (a) The force exerted by gear on shaft 1 $\Rightarrow$

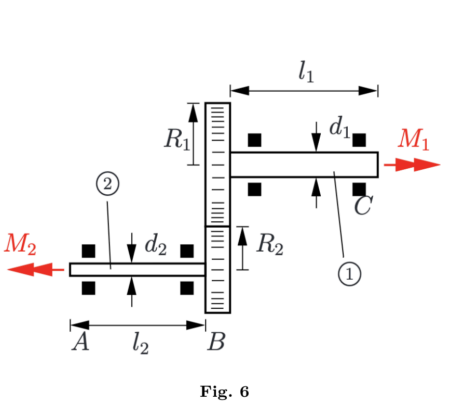


Fig. 6

$$F = \frac{M_1}{R_1}$$

For system to be in equilibrium, the same force must create balancing torque M_2 on shaft 2,

$$M_2 = -FR_2$$

$$\Rightarrow\ \boxed{M_2 = \frac{-M_1}{R_1}\cdot R_2}$$

(b) Given: Admissible shear stress (τ_{admis}).

$$\frac{16M_1}{\pi d_1^3} \le \tau_{admis}$$

$$\rightarrow\ \boxed{d_1 \ge \left(\frac{16M_1}{\pi\tau_{admis}}\right)^{1/3}}$$

Ilrly, for shaft ② $\Rightarrow$

$\frac{16M_2}{\pi d_2^3} \le \tau_{admis}$ $\qquad M_2 = M_1\frac{R_2}{R_1}$ (considering magnitude)

$$\rightarrow\ \boxed{d_2 \ge \left[\frac{16M_1R_2}{\pi R_1\tau_{admis}}\right]^{1/3}}$$

(c) Given: shaft ② is at fixed position A $\Rightarrow$

$$\phi_B = \frac{M_2l_2}{GJ_2}$$

Now, this rotation is transferred to shaft 1 through gears.

Using geometric compatibility $\Rightarrow$

$\phi_1R_1 = -\phi_BR_2$ $\qquad$ ($\phi_1 \Rightarrow$ twist at left end of shaft)

$$\Rightarrow\ \phi_1 = -\frac{M_2l_2}{GJ_2}\times\frac{R_2}{R_1}$$

Twist that shaft ① experiences on its own $\Rightarrow$

$$\phi_{twist} = \frac{M_1l_1}{GJ_1}$$

$\therefore$ Total twist at C $\Rightarrow$

$$\phi_C = \phi_1 + \phi_{twist}$$

$$\phi_C = \frac{-M_2l_2}{GJ_2}\left(\frac{R_2}{R_1}\right) + \frac{M_1l_1}{GJ_1} \qquad \because M_2 = -M_1\frac{R_2}{R_1}$$

$$\phi_C = \frac{M_1l_2}{GJ_2}\times\left(\frac{R_2}{R_1}\right)^2 + \frac{M_1l_1}{GJ_1}$$

$$\phi_C = \frac{M_1}{G}\left[\frac{l_2}{\frac{\pi}{32}d_2^4}\times\left(\frac{R_2}{R_1}\right)^2 + \frac{l_1}{\frac{\pi}{32}d_1^4}\right]$$

$$\boxed{\phi_C = \frac{32M_1}{G\pi}\left[\frac{l_2}{d_2^4}\left(\frac{R_2}{R_1}\right)^2 + \frac{l_1}{d_1^4}\right]}$$